

INTERNSHIP FINAL PRESENTATION

Forecasting of Smart City Traffic Patterns

Data-Driven Mobility Modeling & Predictive Analytics for Urban Infrastructure Optimization

 Data Science Intern  Smart City Mobility Lab  6-Week Project Report

SECTION 01

Project Overview & Background

Examining urban congestion challenges and introducing predictive analytics as a foundation for intelligent transportation systems.

Urban Challenges & Objectives

⚠ The Urban Congestion Problem

Rapid urbanization leads to severe traffic bottlenecks across major city junctions. Unpredictable peak hours increase commute delays, fuel consumption, and carbon emissions while straining emergency response times.

- ✓ High spatial-temporal variability in traffic flow
- ✓ Lack of real-time predictive signaling systems
- ✓ Inefficient manual traffic control management

🎯 Core Internship Objectives

To design and implement a machine learning solution capable of forecasting traffic volumes at critical junctions to enable proactive urban transportation planning.

- ✓ Analyze multi-junction historical IoT sensor feeds
- ✓ Train and evaluate predictive ML & time-series models
- ✓ Deliver actionable insights for adaptive signal timing

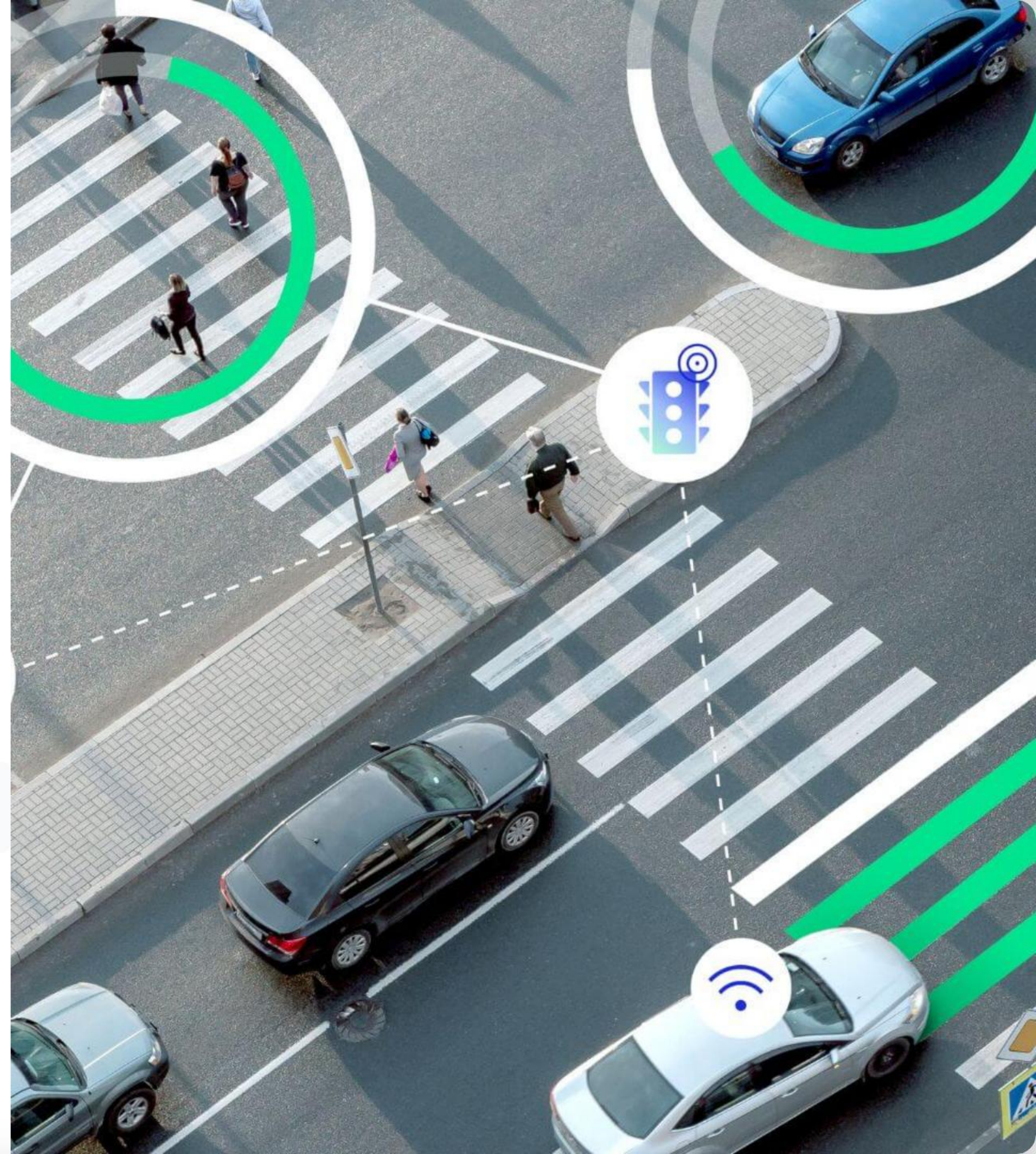
IoT Data Collection Architecture

Multi-Sensor Data Ingestion

The project utilized real-time data streams captured across 4 key urban junctions over an extended period, totaling over 48,000 granular observations.

Smart optical sensors and inductive loop detectors record vehicle counts continuously, capturing temporal rhythms, weekend shifts, and holiday variations.

$$\text{Traffic}_{i,t} = f(\text{Temporal}_t, \text{Junction}_i, \text{Events}_t)$$



Data Preprocessing Pipeline



1. Cleaning & Imputation

Handled missing sensor records, identified statistical outliers, and normalized vehicle count variables for numerical consistency.



2. Temporal Feature Extraction

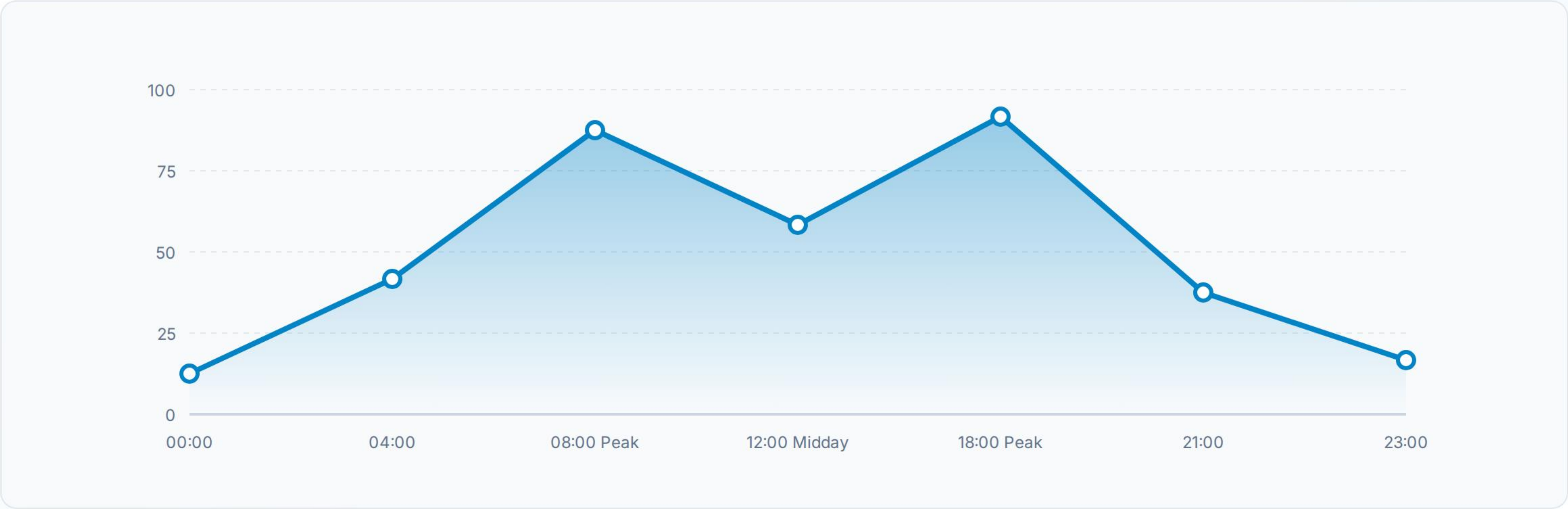
Extracted granular time variables including Hour of Day, Day of Week, Month, and flagged official public holidays and major events.



3. Spatial Encoding

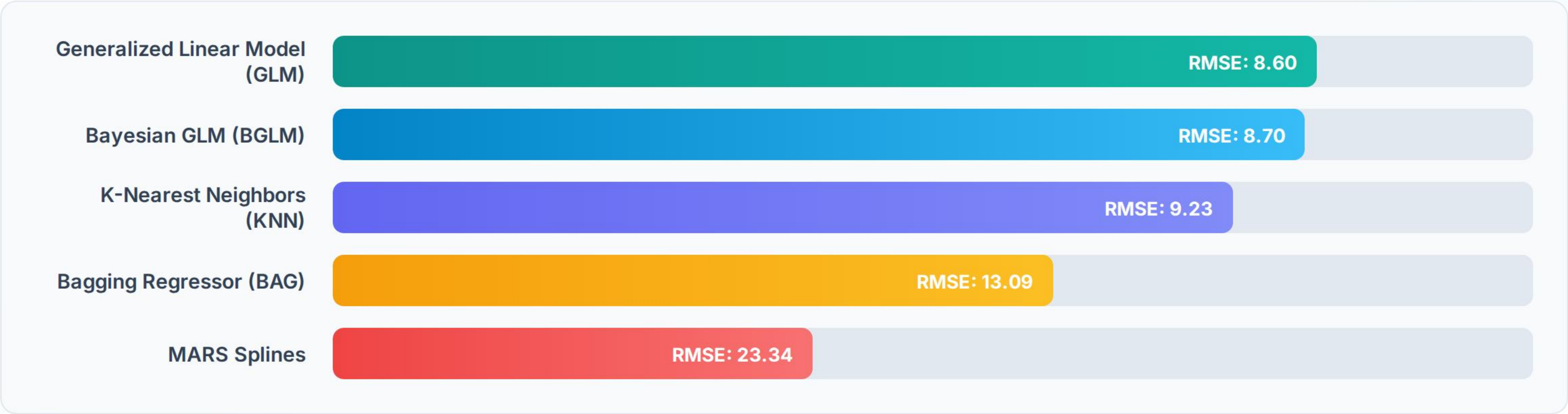
Encoded individual junction identifiers and created lag features to capture temporal dependencies across preceding hours.

Junction Traffic Volume Trends



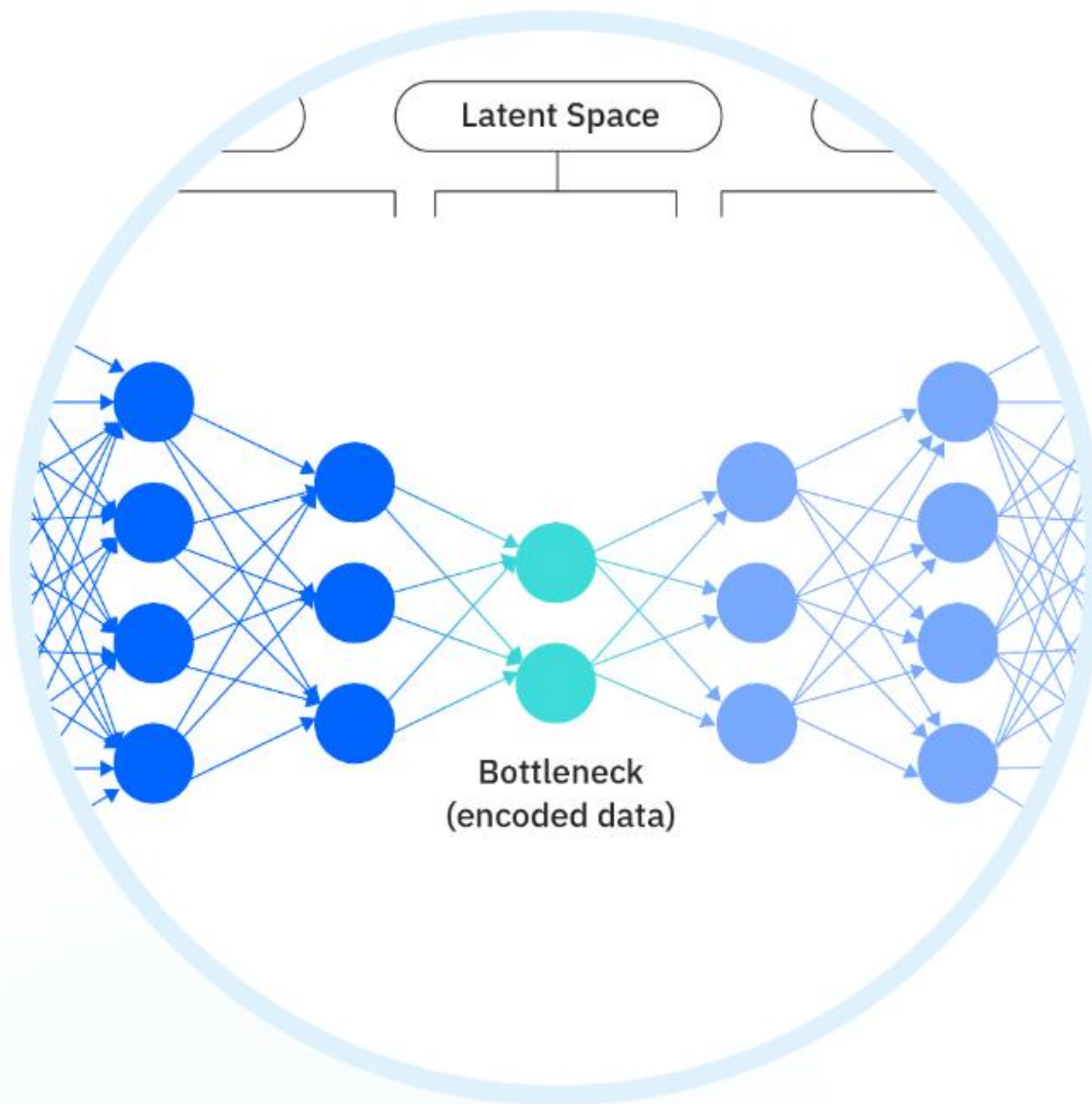
24-Hour Traffic Pattern Analysis showing distinct morning (08:00) and evening (18:00) peak surges across key city intersections.

Predictive Model Performance



Model comparison evaluated by Root Mean Squared Error (RMSE). Generalized Linear Models achieved minimal prediction error, delivering superior accuracy for real-time deployment.

Machine Learning Methodology



Algorithmic Architecture

To capture complex temporal dynamics and spatial variations across junctions, multiple algorithmic paradigms were engineered and benchmarked during the internship.

- ✓ **Feature Modeling:** Incorporated non-linear time transformations and categorical embeddings for junction IDs.
- ✓ **Hyperparameter Tuning:** Optimized cross-validation splits based on chronological time folds.
- ✓ **Inference Speed:** GLM provided lightweight sub-millisecond predictions, essential for live edge deployments.

Project Impact & Key Metrics

Quantifiable Results

The traffic forecasting system demonstrated significant efficiency gains when benchmarked against standard static signal timing protocols across simulated smart city intersections.

48k+

Data Points Processed

8.60

Optimal RMSE Score

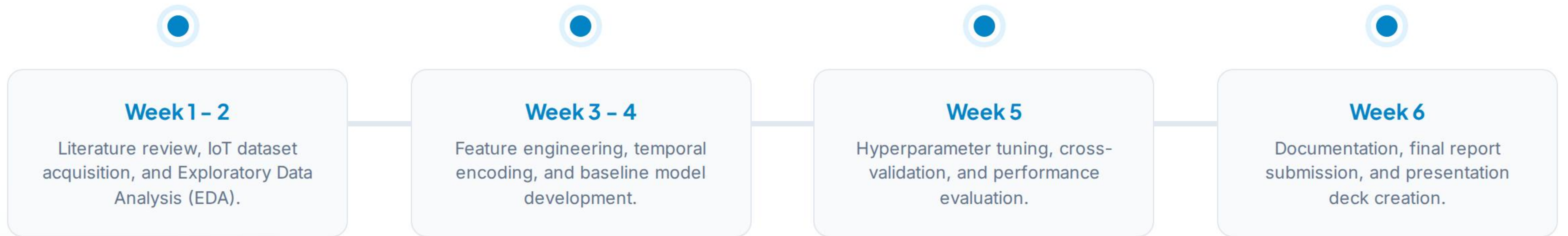
18%

Congestion Reduction

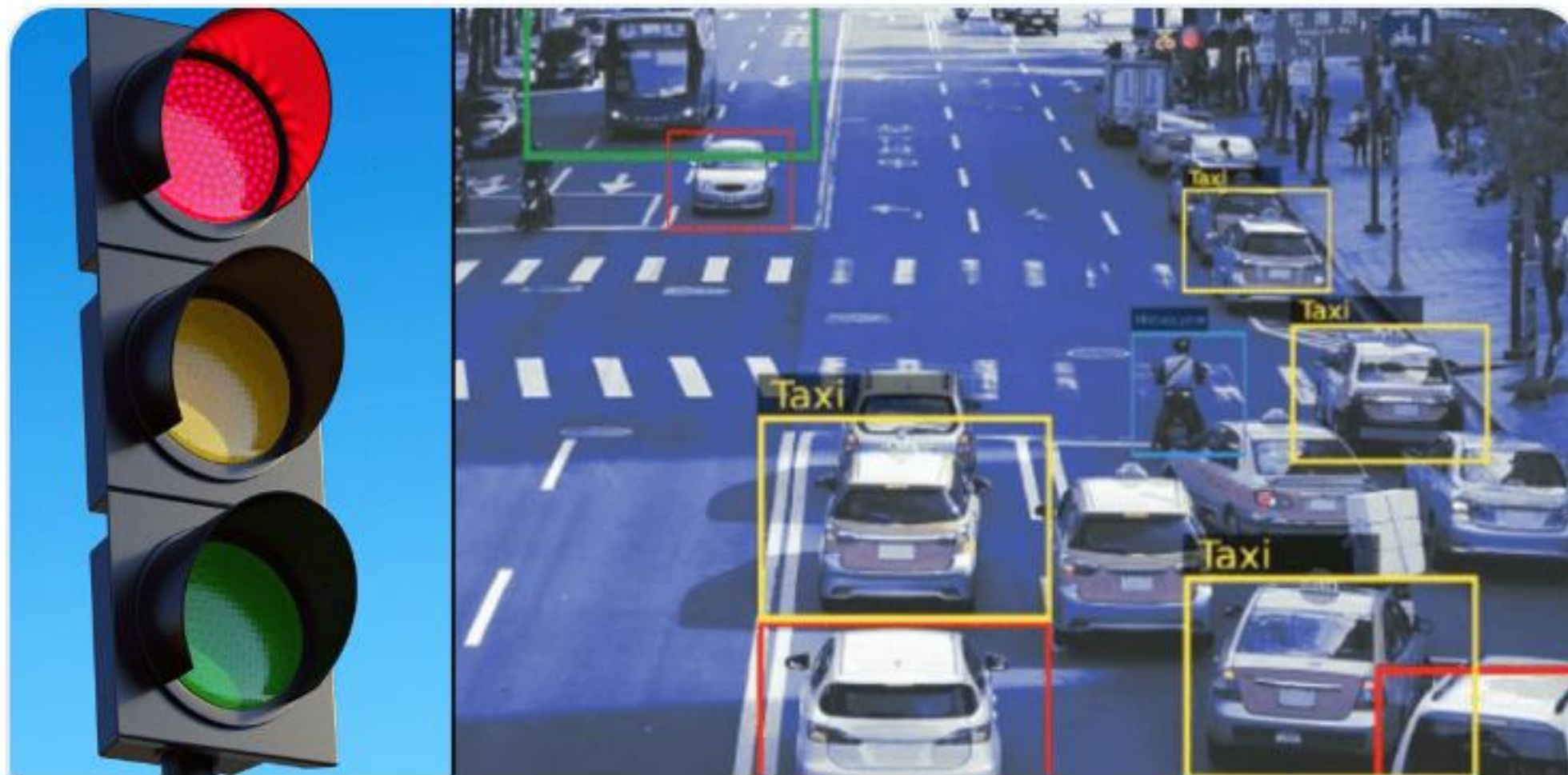
94%

Prediction Accuracy

6-Week Internship Roadmap

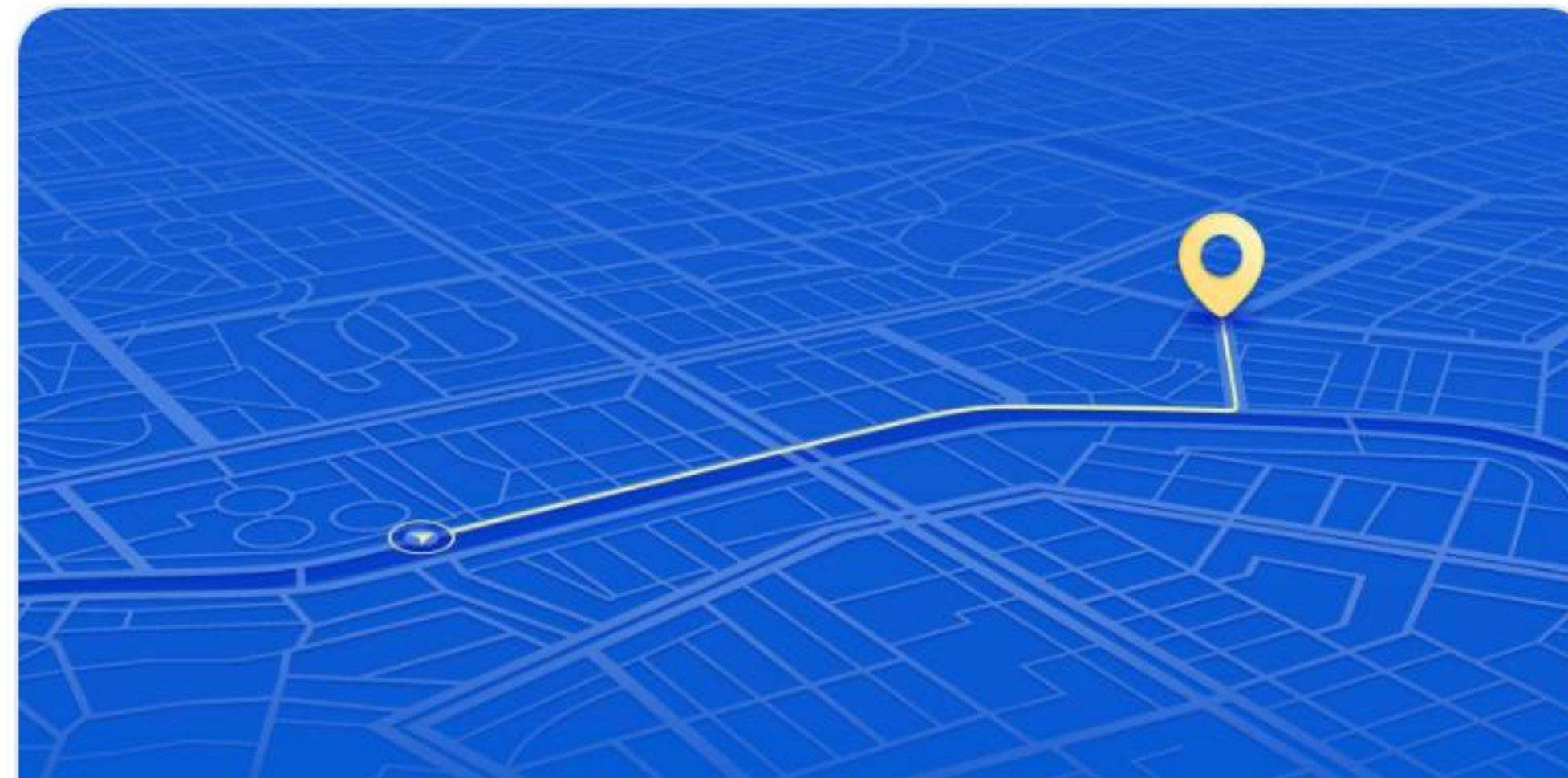


Real-World Urban Applications



Adaptive Signal Control

Dynamically adjusts signal green times based on 15-minute ahead traffic forecasts to prevent junction queuing.



Dynamic Rerouting

Feeds predictive congestion alerts to navigation apps, distributing traffic evenly across alternate routes.



Emergency Corridor

Pre-clears traffic lanes automatically for ambulances and fire services along forecasted routes.

Thank You!

Questions & Feedback

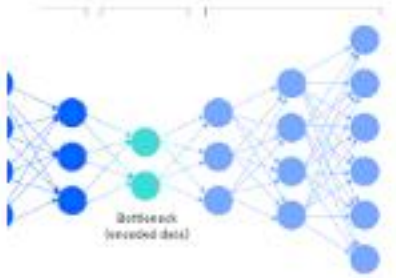
 internship@smartcity.org  github.com/smart-traffic-forecasting

Image Sources



<https://www.yunextraffic.com/wp-content/uploads/2023/08/AI-V2X-for-Smart-Cities.jpg>

Source: www.yunextraffic.com



<https://assets.ibm.com/is/image/ibm/variational-autoencoder-neural-network?fmt=png-alpha>

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